

PUBH 526

Topic 1, Part 1:
Introduction and Review

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Introductions

- Your name
- What degree(s) are you working on?
- What is one application of statistics that you've found especially cool?

About This Course

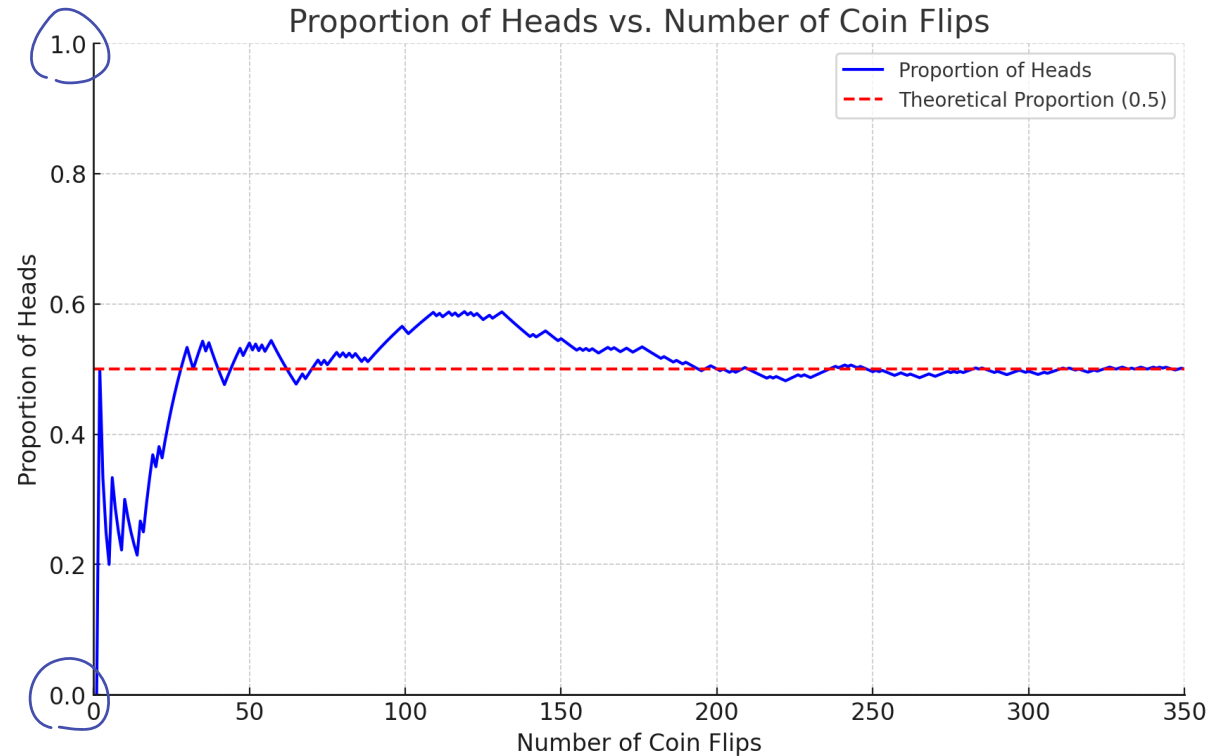
- This is a statistics course
- Please make sure to read the syllabus completely, and ask any questions
- Additional details can be found on Brightspace
- We will rely on Brightspace a lot for communication, assignments, etc. ... please check site often

Probability

- Randomness ... uncertainty, but with known (or assumed) structure

- Long-run or limiting frequency *Approaching theoretical ex: 0.5*

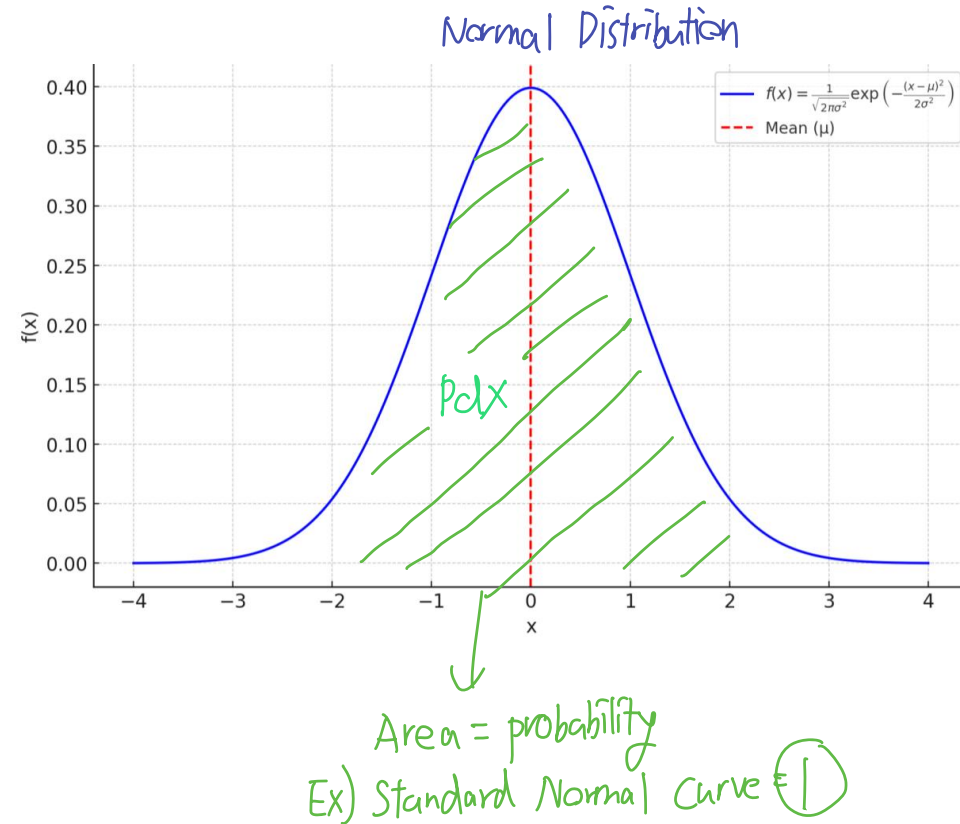
- What are some properties of probabilities?
 $0 < P(E) < 1$
Impossible $\rightarrow$ *Certain*



- ① $[0, 1]$
 $\downarrow$
Impossible $\rightarrow$ *Certain*
- ② Entire set of possible outcomes = Always 1.
- ③ Prob of Event + No Event = Also 1.
(Complement)

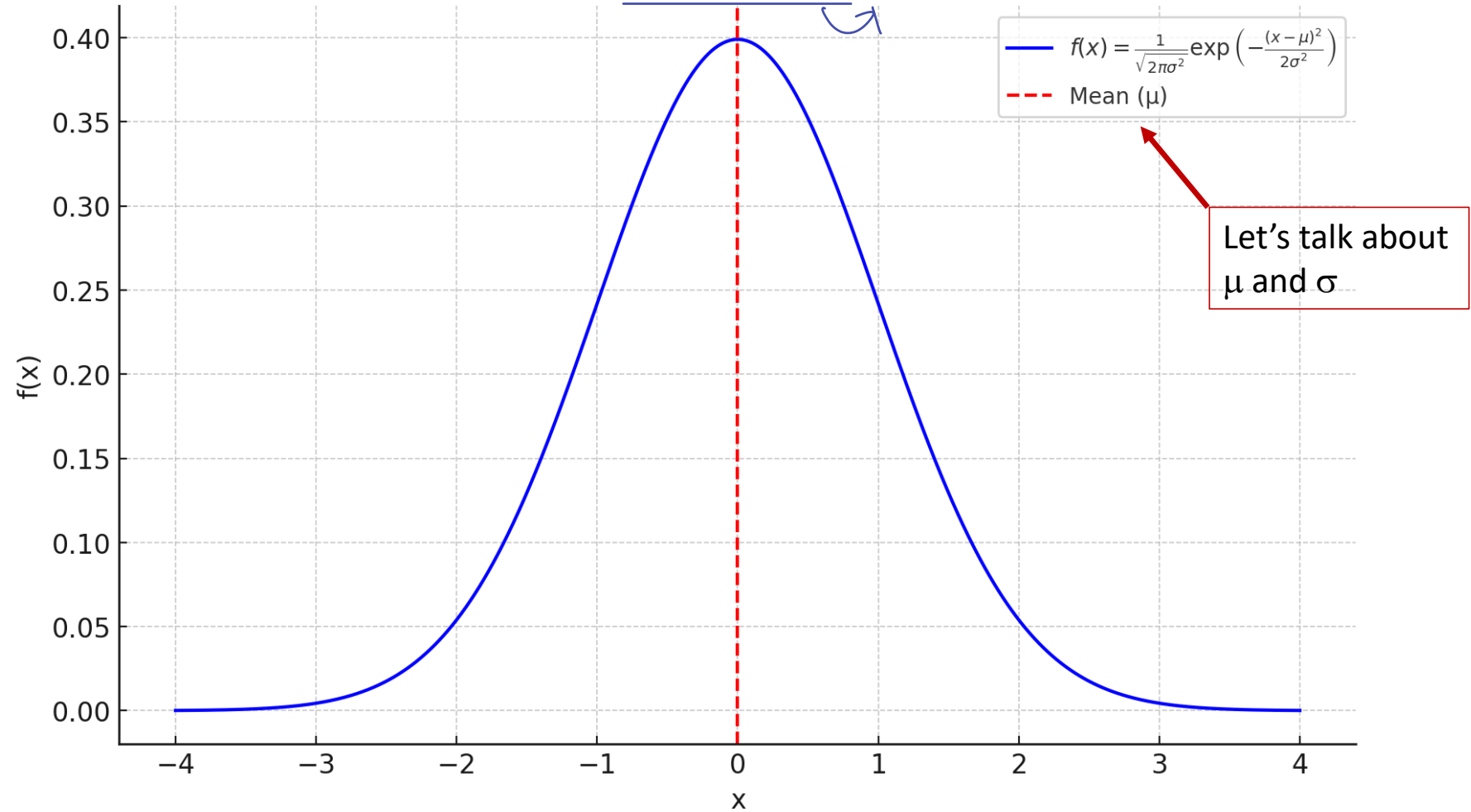
Random Variable

- The **outcome** of a process involving **chance**
 - Often denoted with a capital letter, e.g., X
- Before this process happens, there are different possible values, with different probabilities
- Once the process happens, you have one realized value
 - Often denoted with a lower-case letter: $X = x$
- This is all in contrast to a **constant**



Probability Density Function of a Normal Distribution

$(\mu = 0, \sigma^2 = 1)$ *Standard Normal*



Normal Distribution

"Distributed As..."

$$X \sim N(\mu, \sigma^2) \quad (\text{Mean, Variance})$$

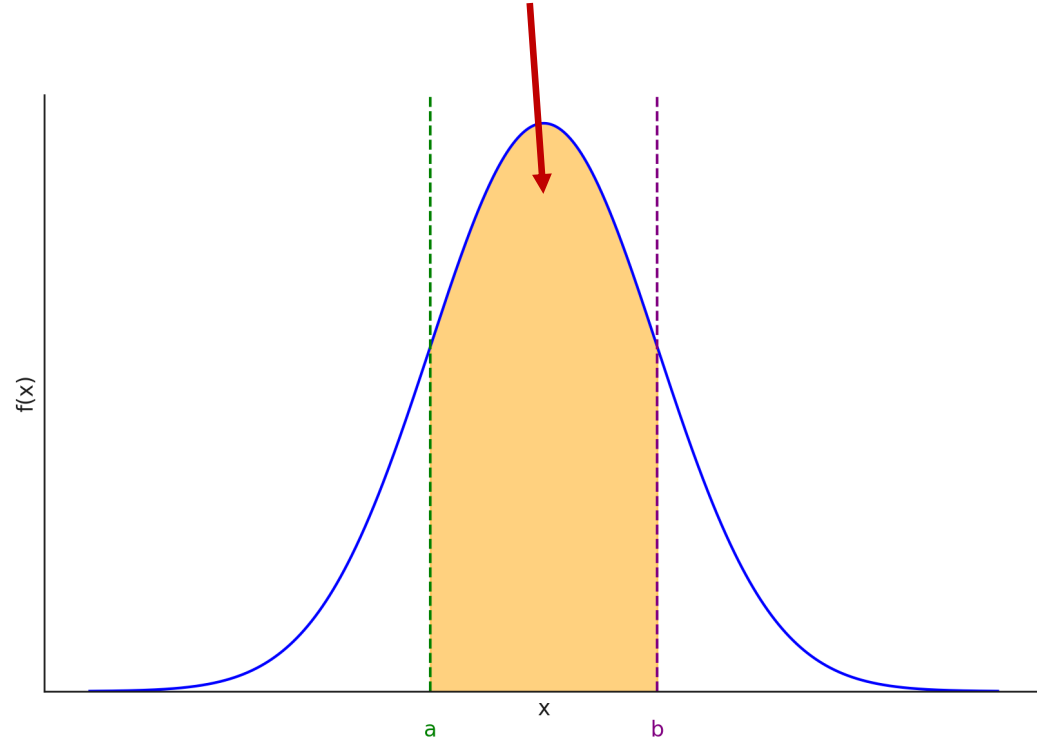


Random Variable

What are the possible x for a normally distributed random variable?

What is the total area under the curve of this PDF? $\text{Sum} = 1$
Probability $[0, 1]$

Area
Probability
This area is $P(a < X < b)$



Expected Value

- The expected value of a random variable, X , is its average, denoted $E(X)$
- Some random variables do not have an expected value! *Depends on whether the mean exists*
- If $X \sim N(\mu, \sigma^2)$, then what is $E(X)$? Average
 $E(X) = \mu$

parameter: unknown constant
statistic: unknown estimate of parameter
sampling distribution: distribution of statistic $E(x) \bar{X}$

what's a bad estimate?
↳ outliers
↳ random noise
↳ bias ... etc.